

Build your Own Lantern Robot



Lantern | **Material List:** Check before start

3D PRINTED COMPOMENTS

- [ ] Housing Base
- [ ] Pulley Belt Clamp
- [ ] Main Pulley
- [ ] Pulley Shaft
- [ ] Pulley Cap
- [ ] Pulley Stand
- [ ] Belt Guide
- [ ] E-Component Tray
- [ ] Top Cap
- [ ] Top Brace
- [ ] Top Belt Clamp
- [ ] Base Holder

ELECTRONIC COMPOENTS

- [ ] Servo Motor
- [ ] Breadboard
- [ ] Raspberry Pi Pico
- [ ] Voltage Converter
- [ ] Motor Driver
- [ ] Power Port
- [ ] Vibration Motor
- [ ] RGB LEDs (optional)
- [ ] IMU (optional)

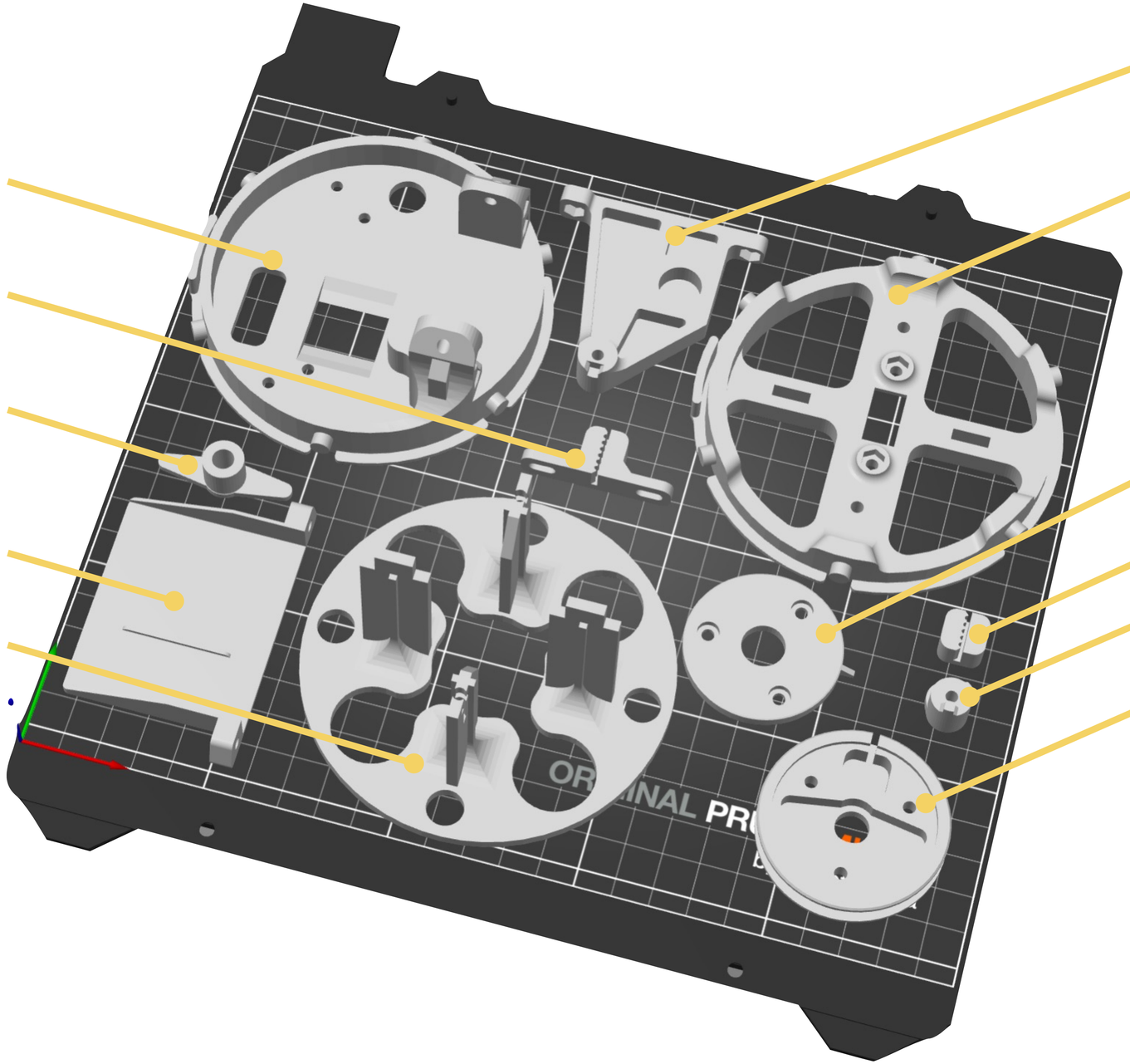
OTHERS

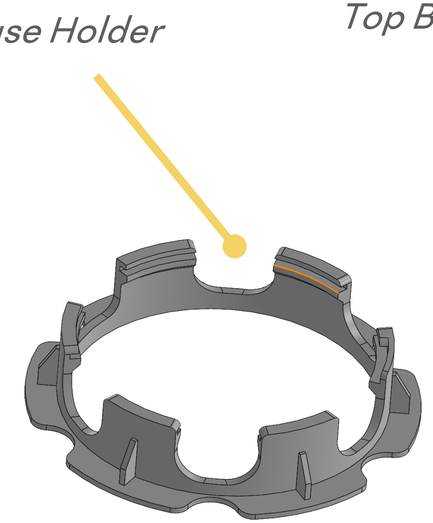
- [ ] GT2 Belt
- [ ] Sliced PETG Sheet
- [ ] Jump Wires
- [ ] Power Adapter
OR
Battery
- [ ] 7x M3*10 Screw
- [ ] 14*M3 Nut
- [ ] Resistant (optional)
- [ ] 7x M3*20 Screw
- [ ] Two Wing Servo Horn
- [ ] Servo Horn Screw (Come with Servo)

TOOLS

- [ ] Screw Driver
- [ ] Double-sided Foam Tape
OR Hot Glue OR Super Glue
- [ ] Pico to PC Connection Cable
- [ ] Soldering Iron (optional)
- [ ] Soldering Accessories (optional)
- [ ] Craft Knife

Lantern | 3D Printed Components

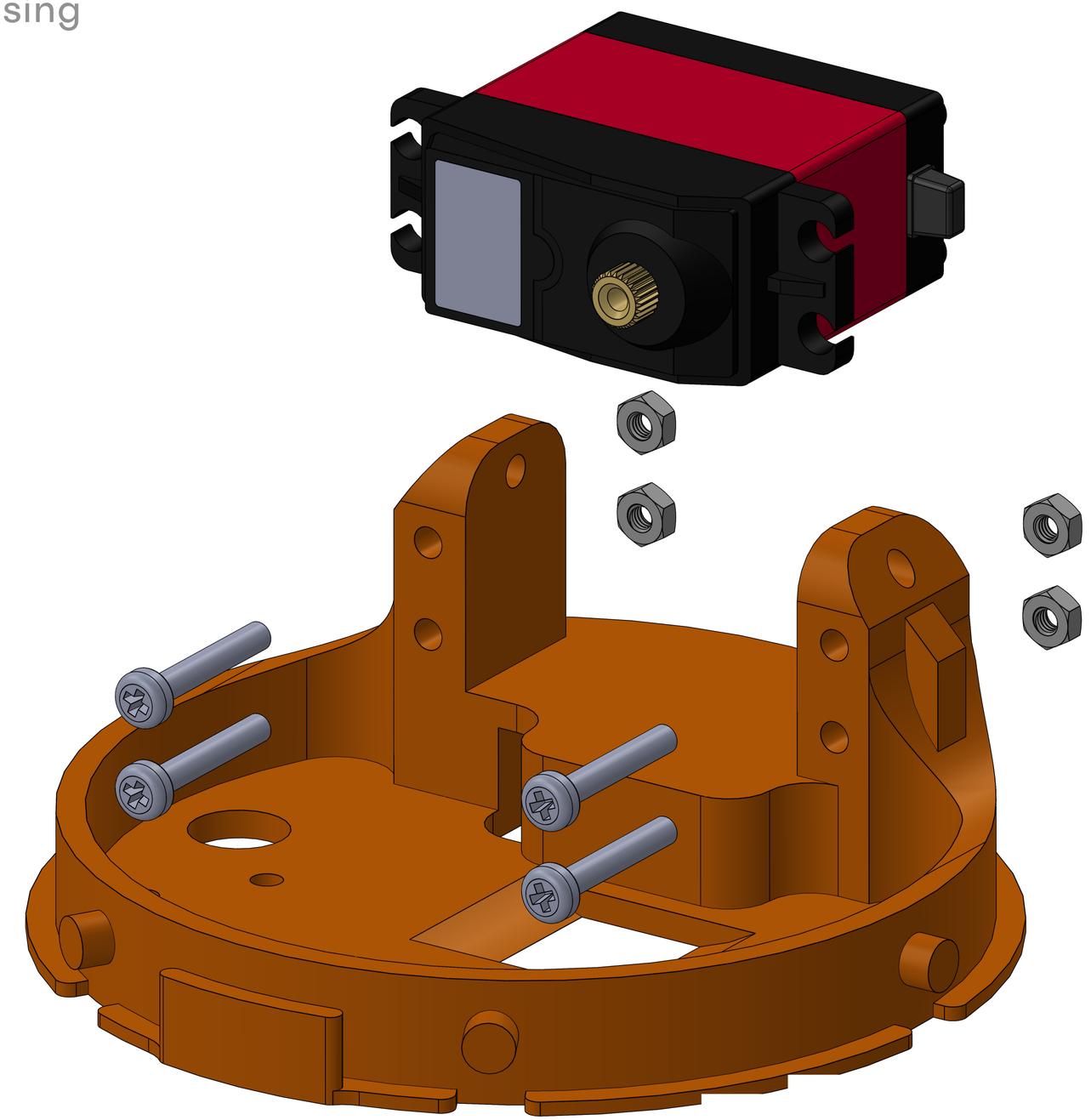




Lantern | **Step 1:** Mount Servo Unit to Base Housing

1. Place the servo into the Base Housing following the direction shown in the visual
2. Secure the Servo Motor with 4x M3*20 screws and 4x M3 nuts

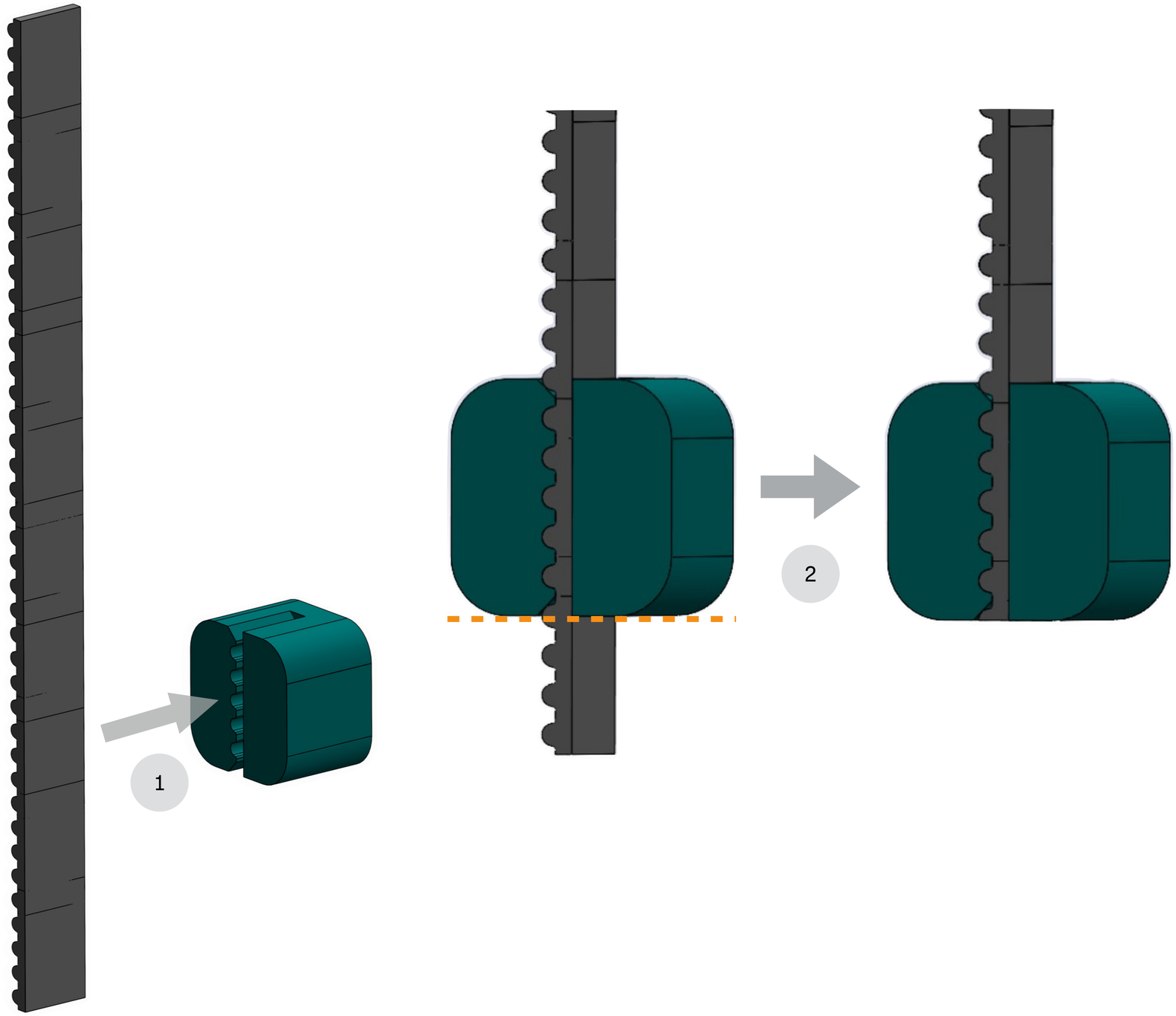
Materials Needed:
Base Housing
Servo Motor
4 M3*20 screws
4 M3 nuts



Lantern | **Step 2:** Prepare belt

1. Cut out around 25 cm of the GT2 Belt
2. Clamp one side into the belt clamp
3. Cut of the vestige portion

Materials Needed:
Pulley Belt Clamp
Belt

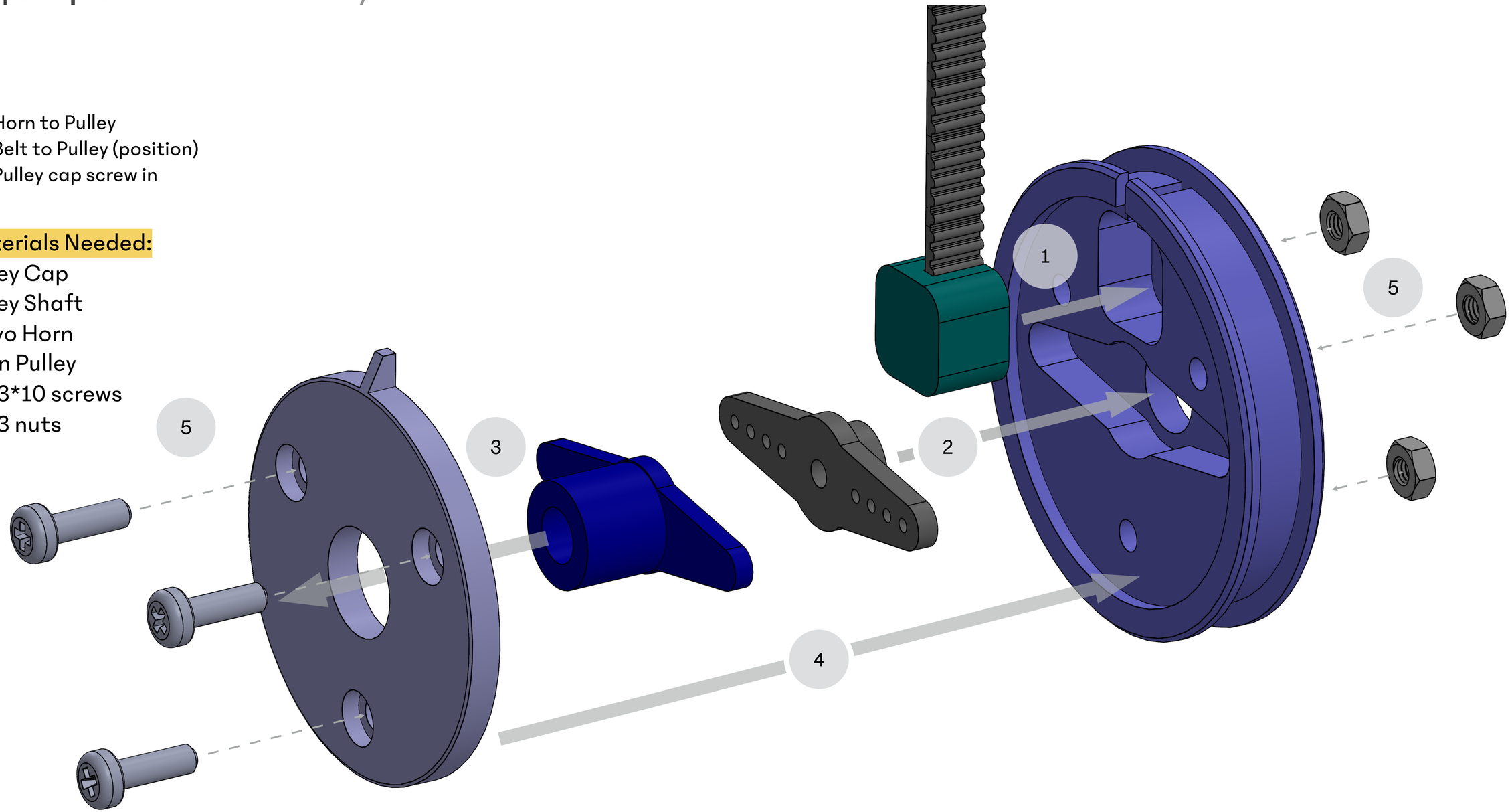


Lantern | **Step 3:** Assemble Pulley

1. Horn to Pulley
2. Belt to Pulley (position)
3. Pulley cap screw in

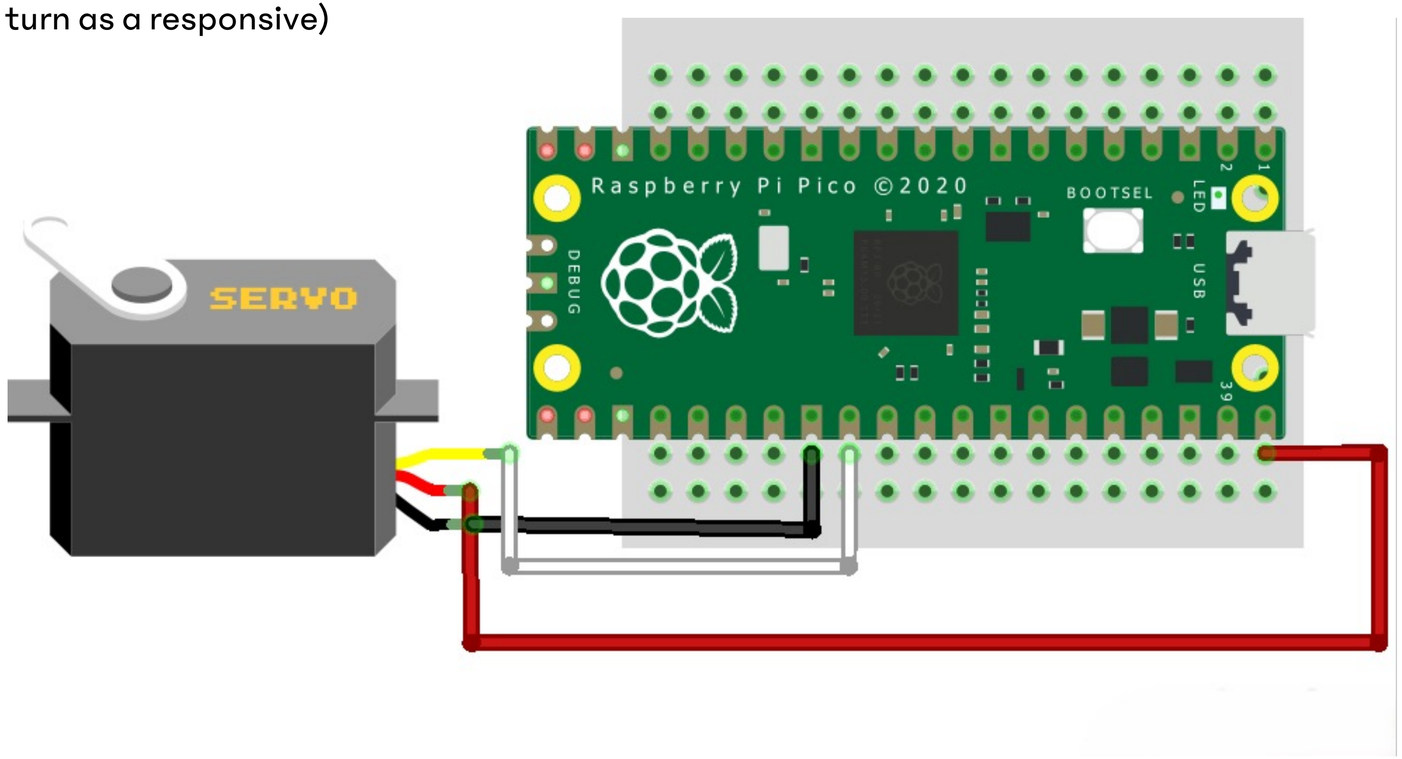
Materials Needed:

Pulley Cap
Pulley Shaft
Servo Horn
Main Pulley
3 M3*10 screws
3 M3 nuts



1. Go to **thonny.org** to download & run **thonny**.
2. Install **micropython-servo** and **picozero** packages.
   This is under Tools -> Manage Packages.
3. Connect the servo to the Raspberry Pi using **this** wiring diagram.
4. **Run** the code below. (the motor should turn as a responsive)

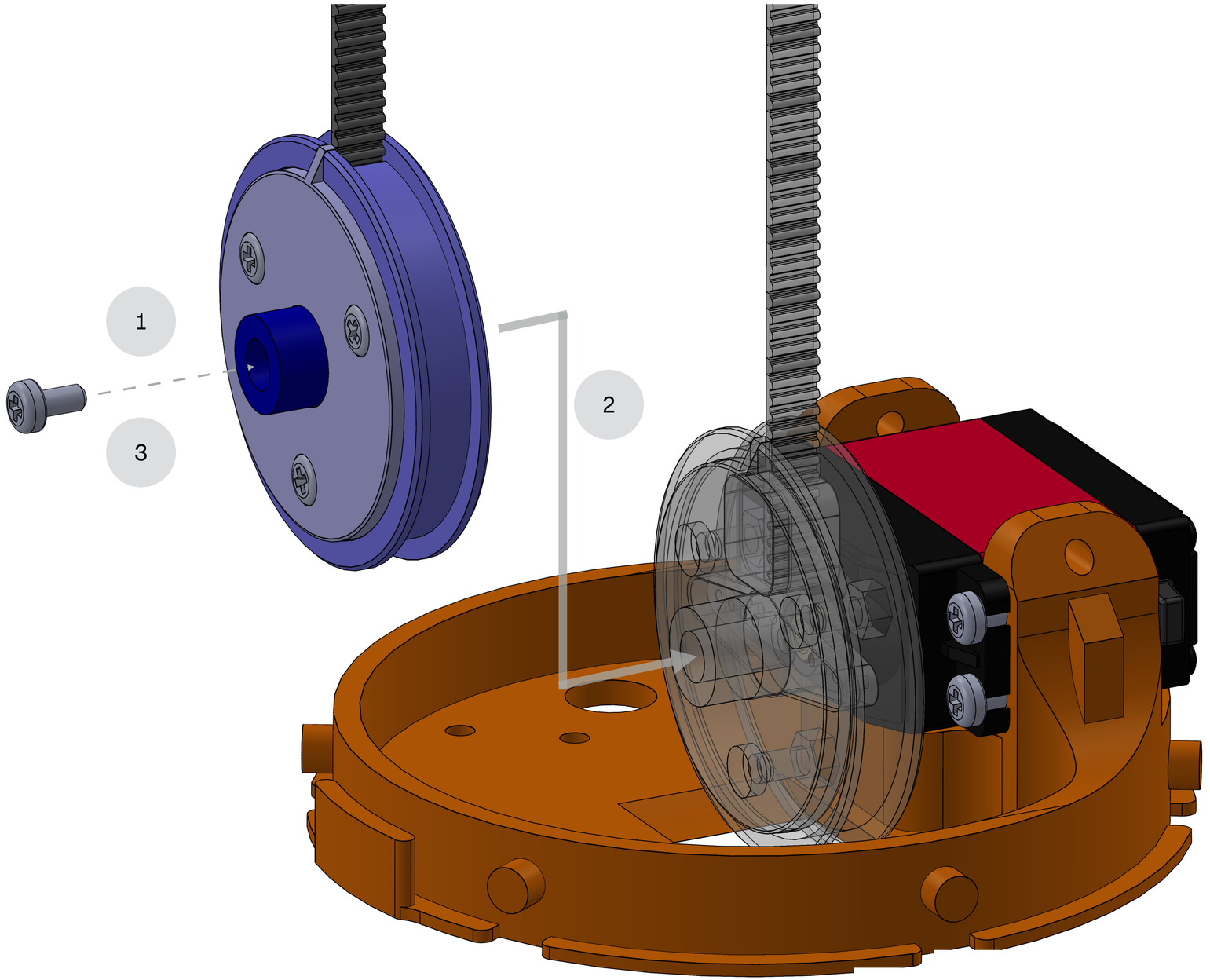
Materials Needed:
Servo
Jump Wires
Raspberry Pi Pico
Micro USB Cable



Lantern | **Step 5:** Assemble Pulley to Servo

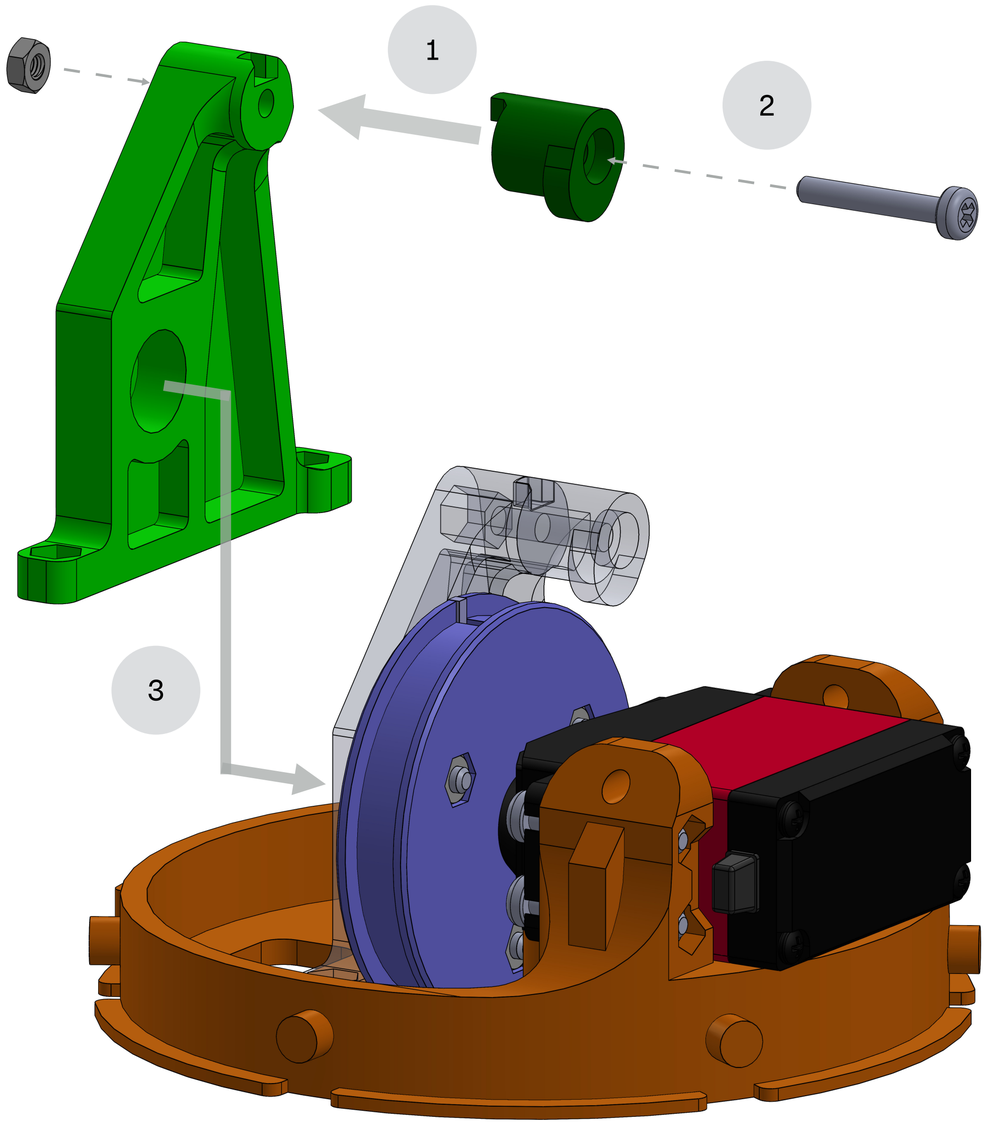
1. Horn Screw in
2. Mount Pulley to Servo
3. Tighten screw

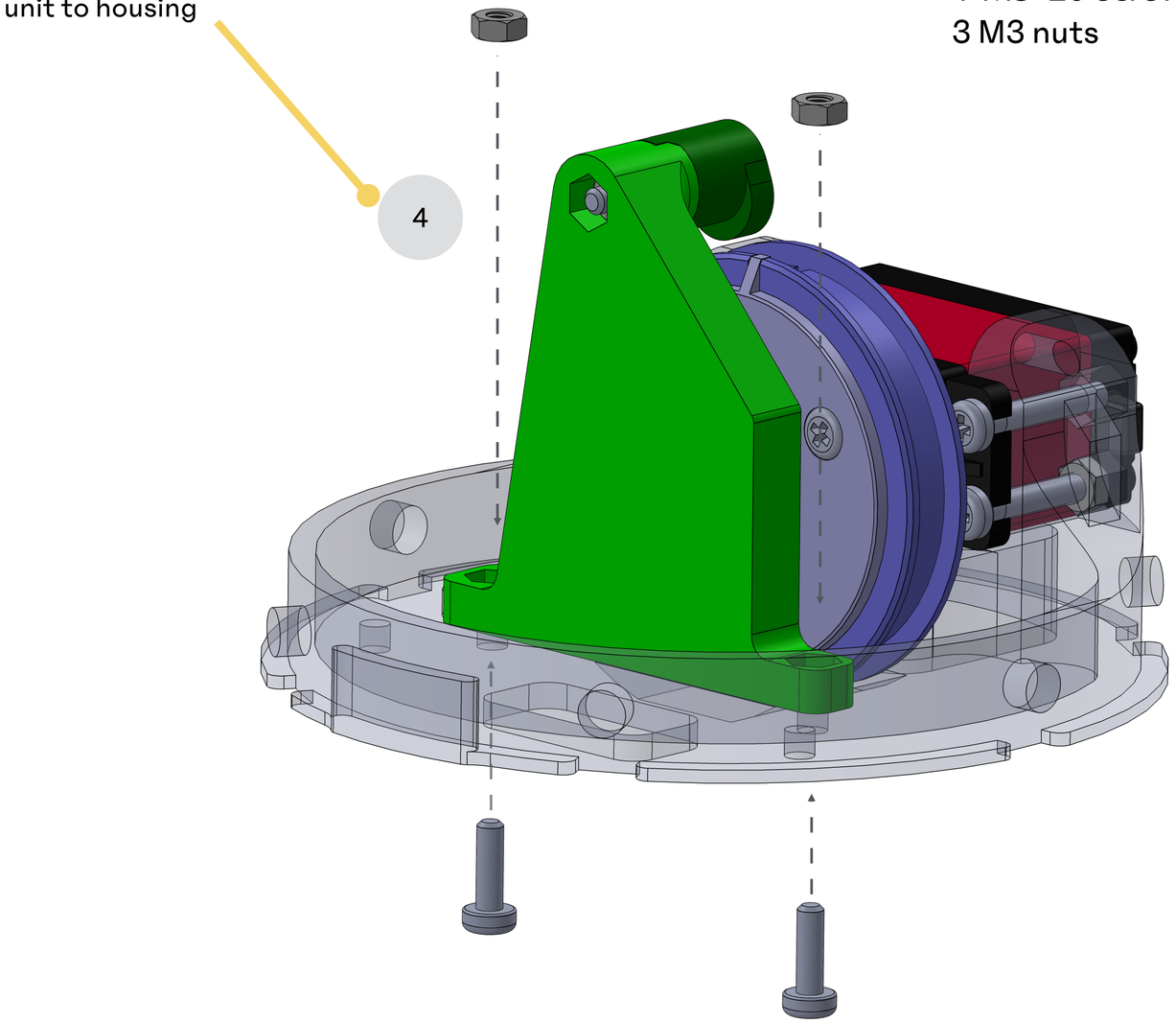
Materials Needed:
M3*10 screws



Lantern | **Step 6:** Mount Belt Guide to Base Housing

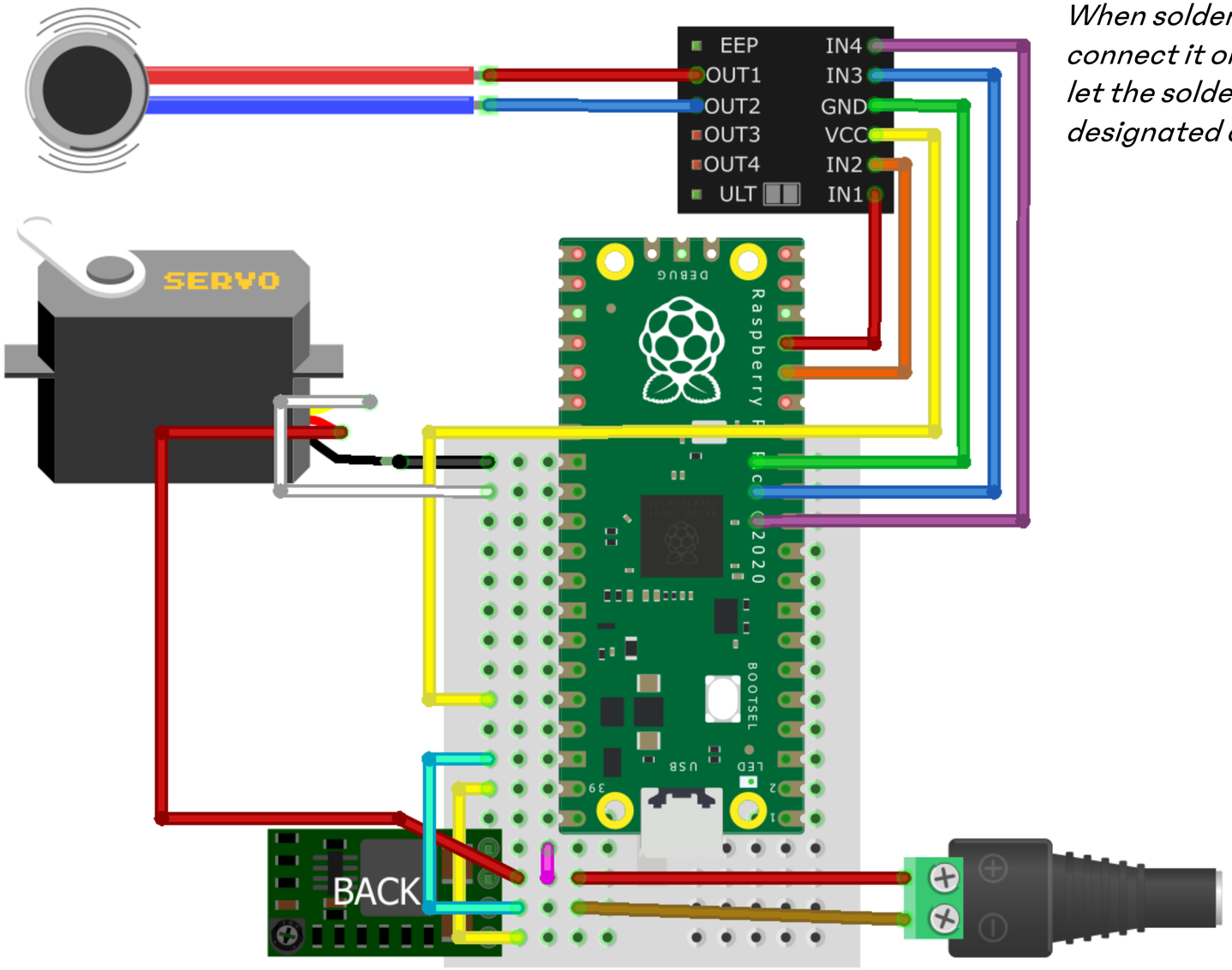
Materials Needed:
2 M3*10 screws
1 M3*20 screws
3 M3 nuts



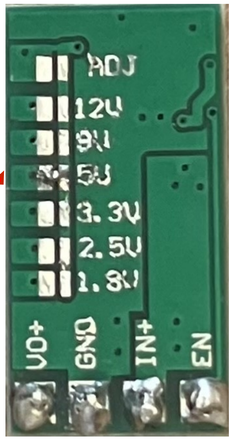


Materials Needed:

- Jump Wires
- Servo Motor
- Breadboard
- Raspberry Pi Pico
- Voltage Converter
- Motor Driver
- Power Port
- Vibration Motor



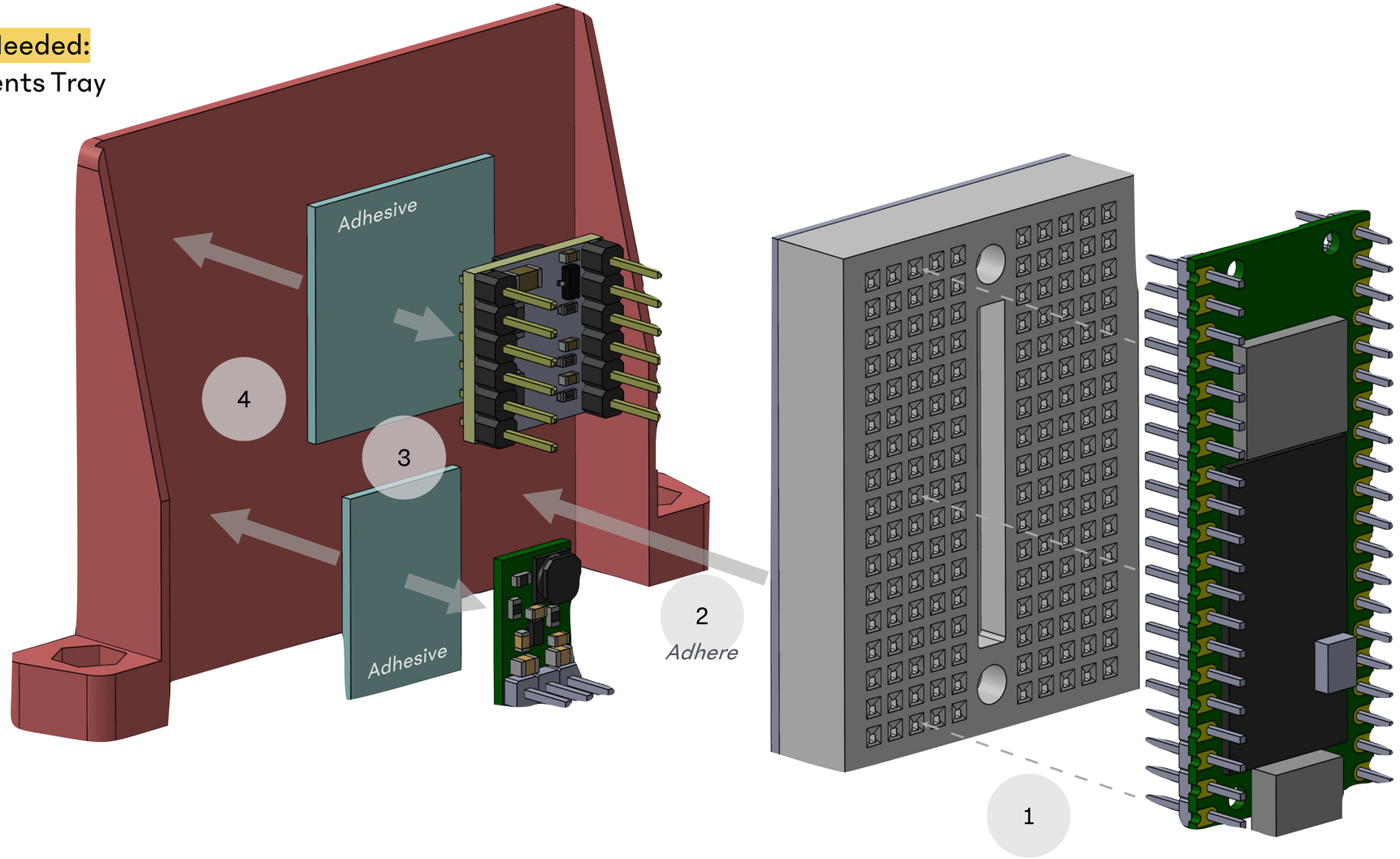
When soldering the voltage converter, make sure to connect it only to the 5V point, and be careful not to let the soldering material spread beyond the designated area.



Lantern | **Step 8:** Mount E-component to E-component Bracket

You can choose double-sided adhesive or hot glue gun

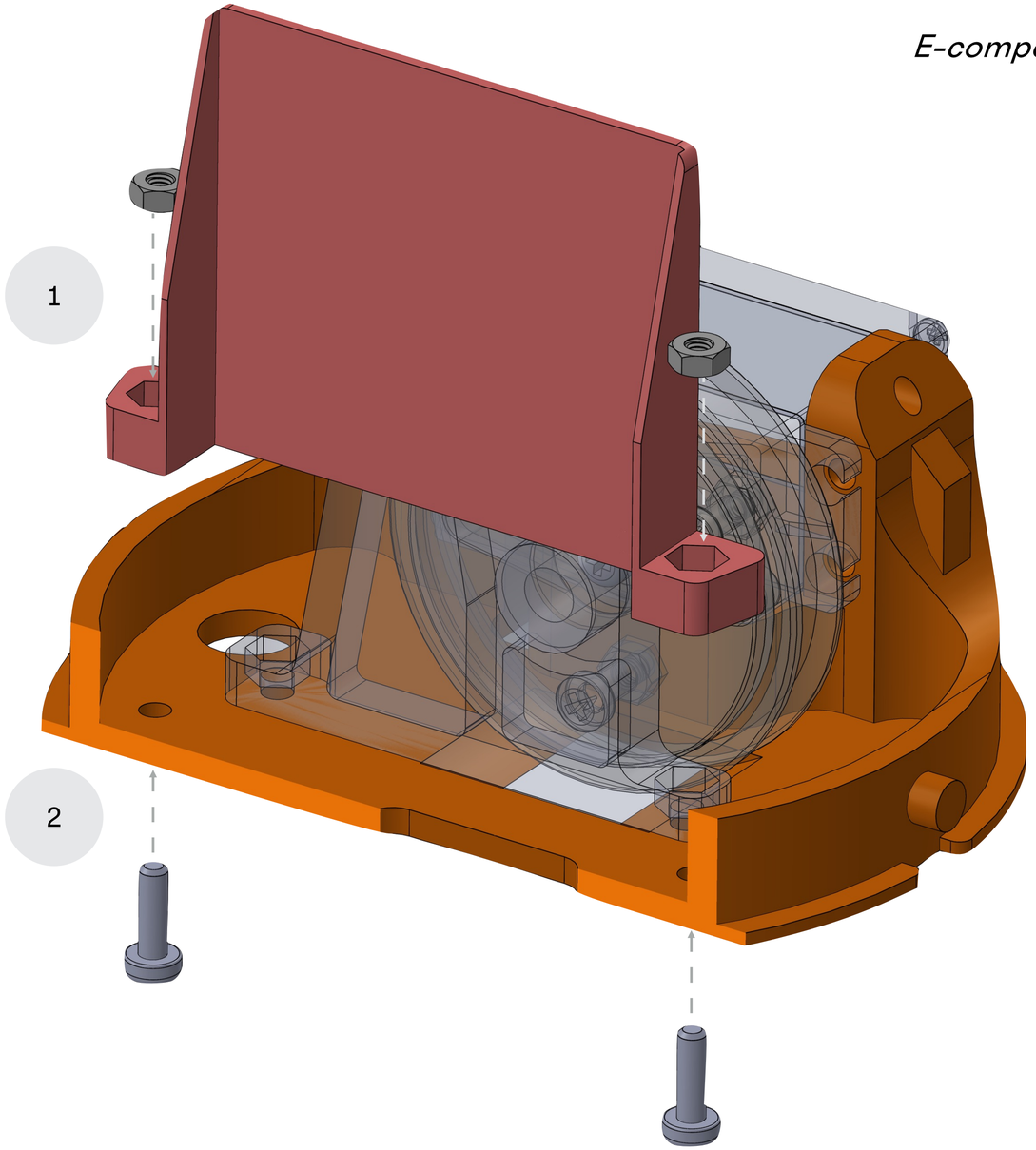
Materials Needed:
E-components Tray



Lantern | **Step 9:** Mount E-component Bracket to Base Housing

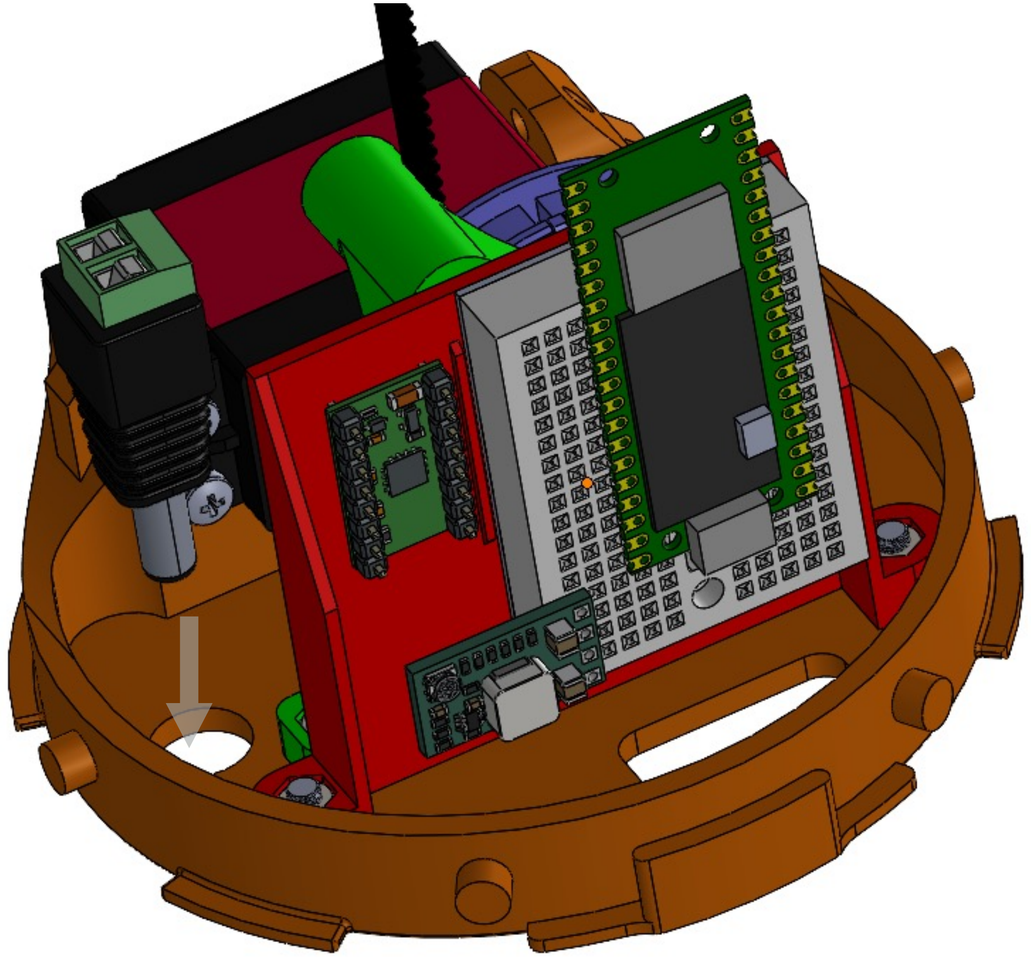
E-components and Wires are hidden for simplicity purpose

Materials Needed:
2 M3*10 screws
2 M3 nuts



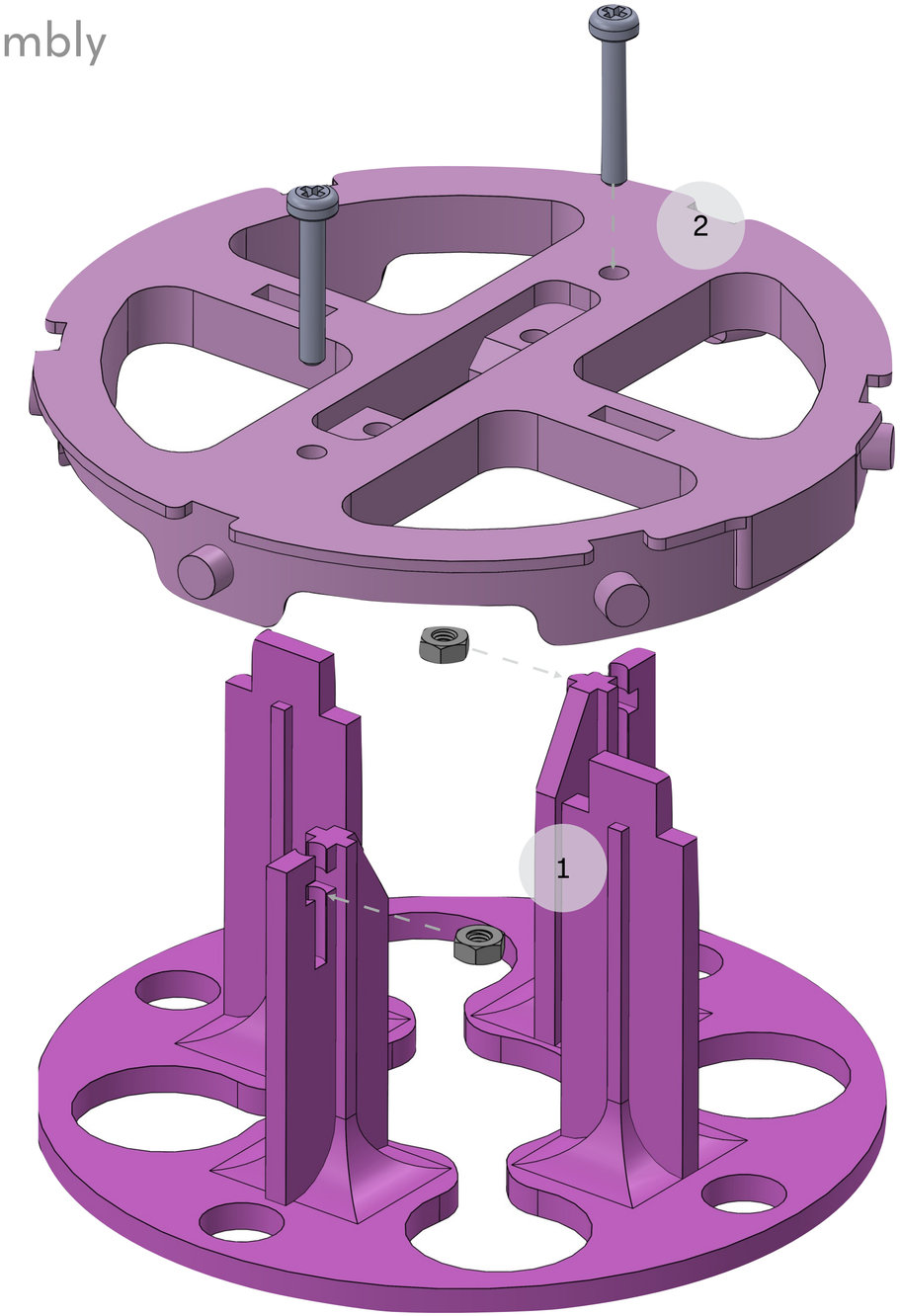
Lantern | **Step 10:** Hot glue Power Port

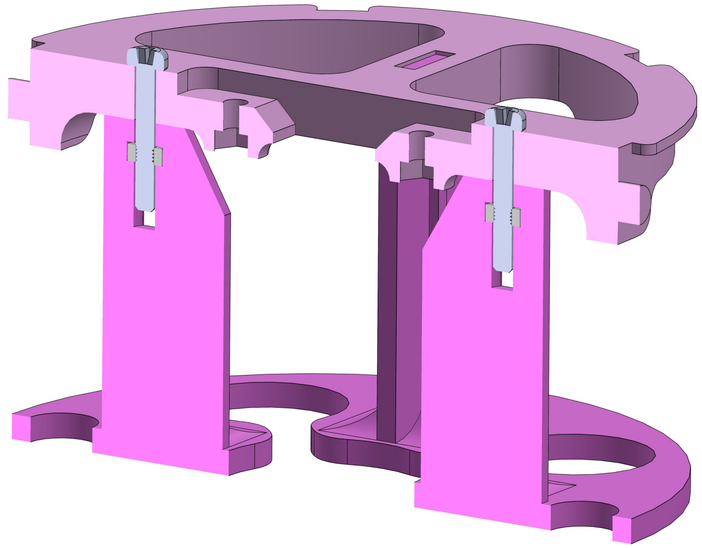
Materials Needed:
Power Port
Glue gun



Lantern | **Step 11:** Top Cap Assembly

Materials Needed:
Top Cap
Top Brace
2 M3*20 screws
2 M3 nuts

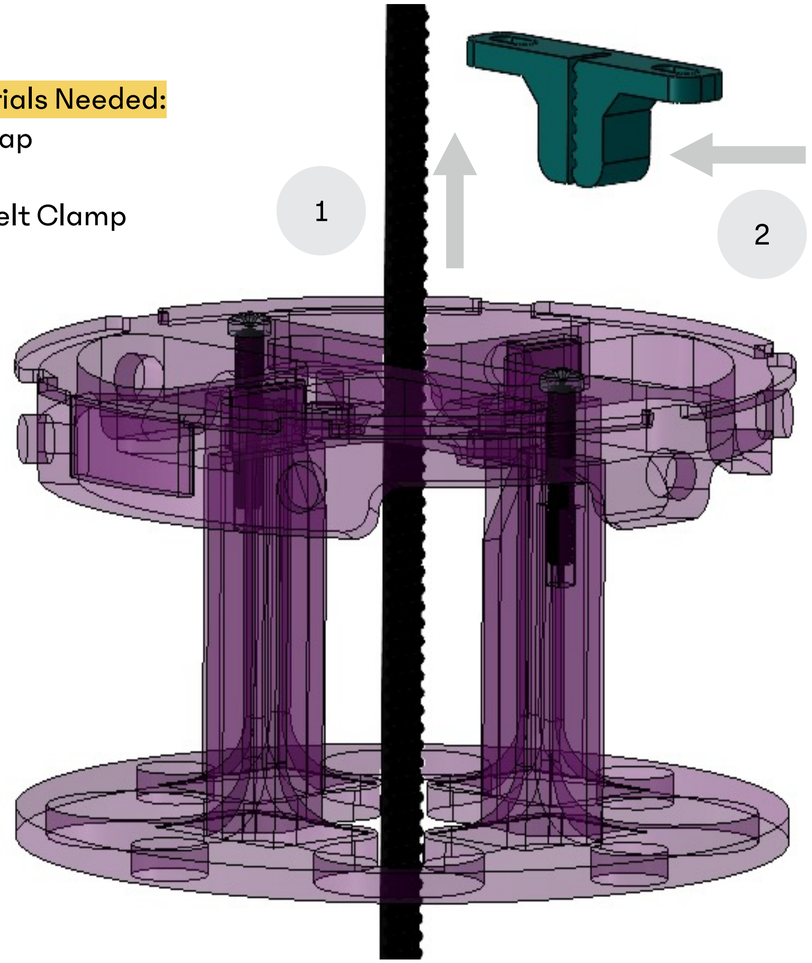


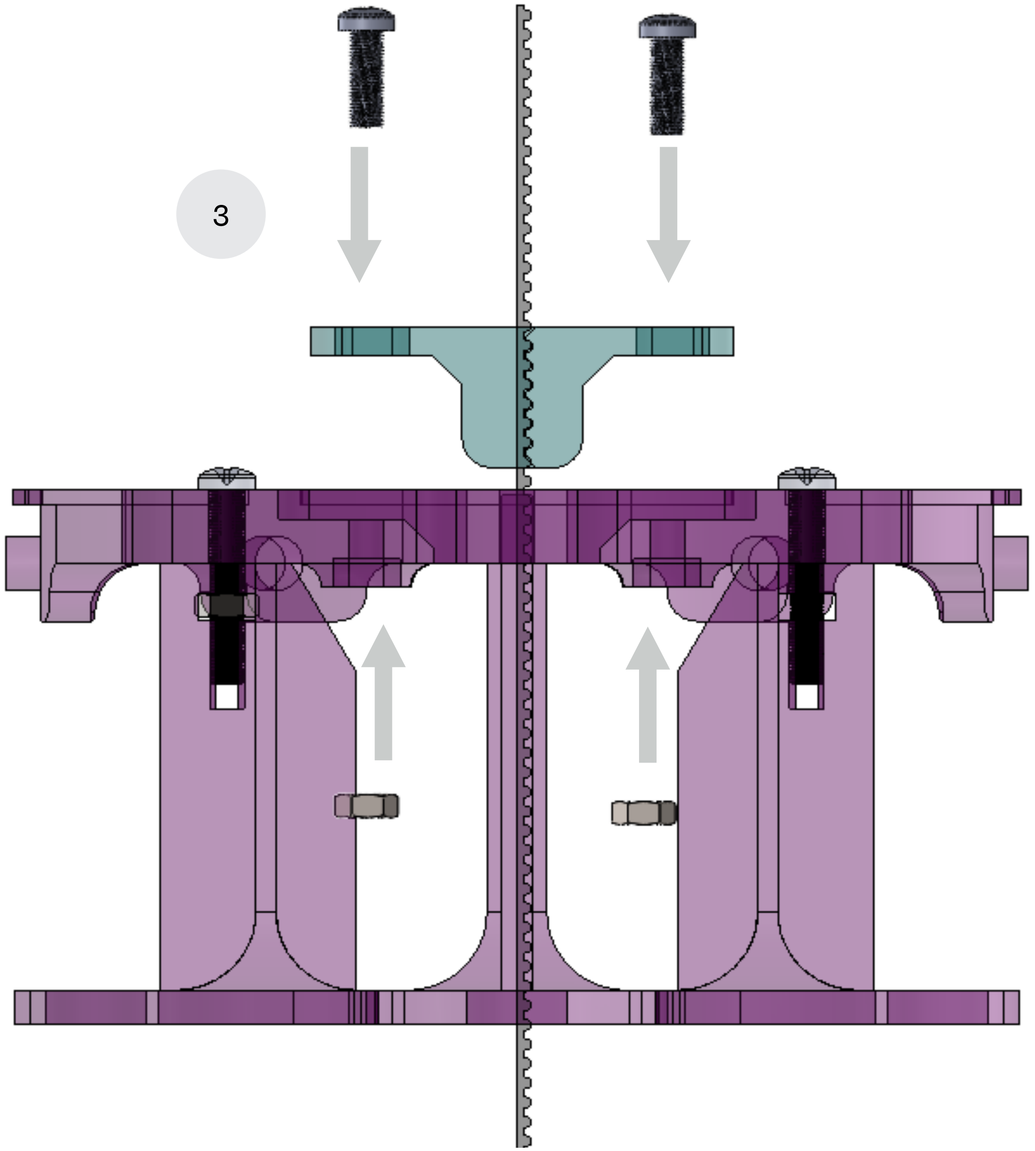


Lantern | **Step 12:** Top Cap Assembly

1. Belt through the Top Cap
2. Put Top belt Clamp into the belt clamp
3. Screw in

Materials Needed:
Top Cap
Belt
Top Belt Clamp

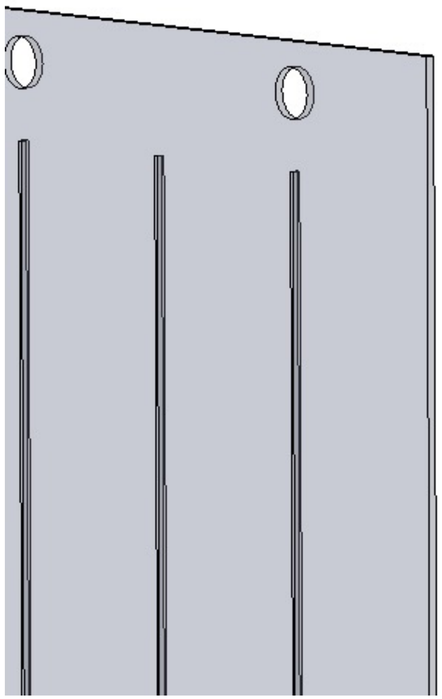


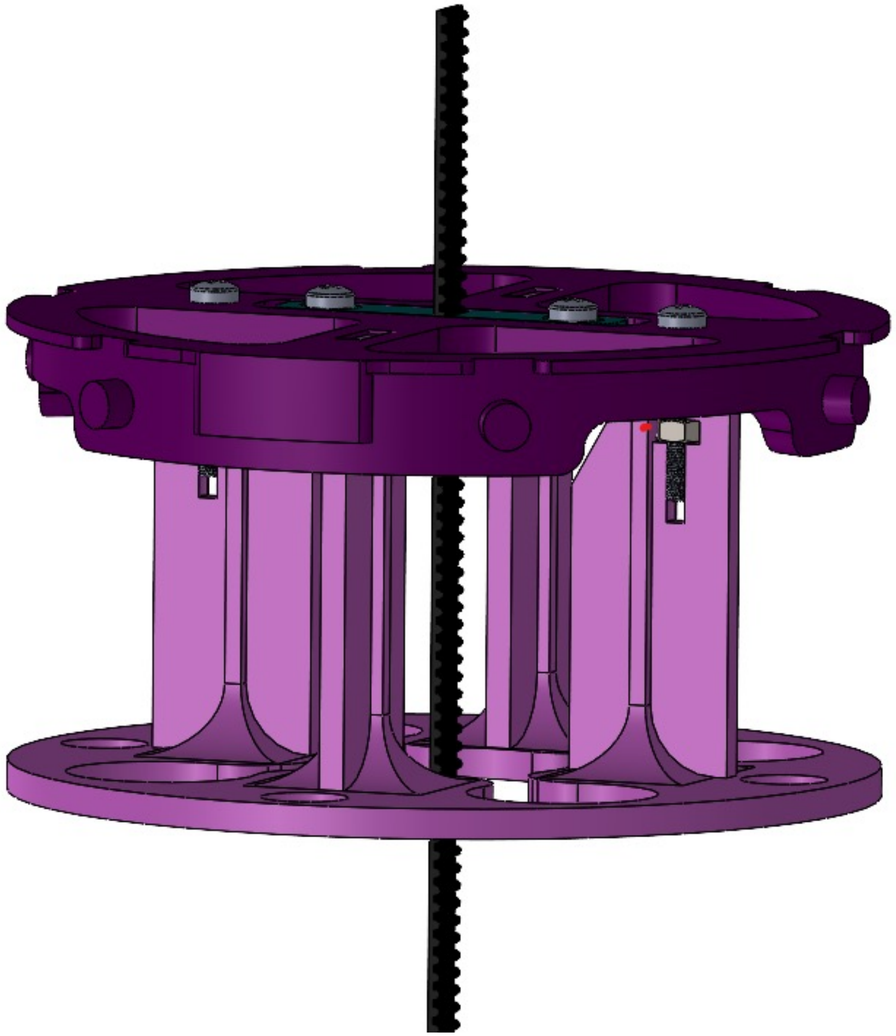


Lantern | **Step 13:** Connect Base Housing and Top Cap with Shell

1. Secure Vibration motor to Top Brace
2. Make sure the shell fits into the slot of the Top Cap. Also, for the Base Housing

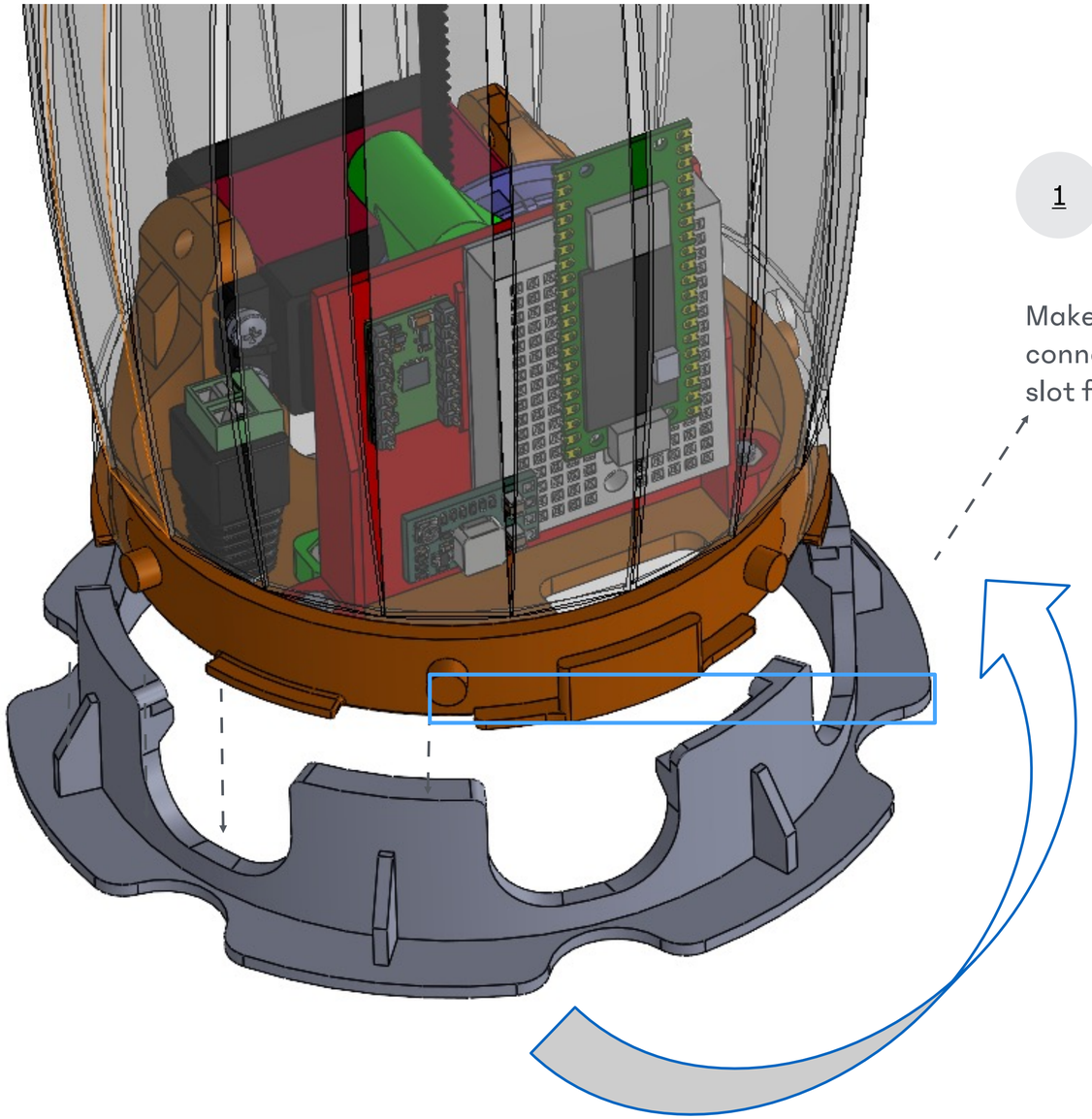
Materials Needed:
Shell





Lantern | **Step 14:** Connect Base Housing and Base Holder

Materials Needed:
Base holder

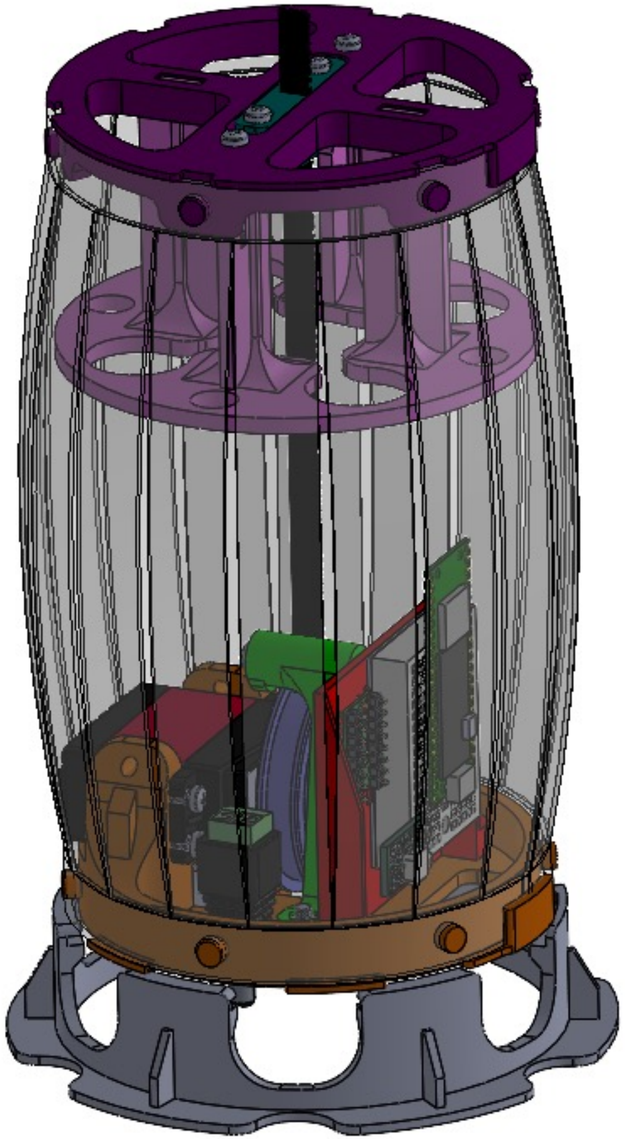


1

Make sure this part is connected to the shorter slot first

2

Spin it until it's no longer spinning



Please ensure that all wires are arranged properly to avoid entanglement with the casing.